Jiwon **Choi**

Robotics Researcher · SLAM & Spatial Perception

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Summary

I work on resilient state estimation for robots that must localize with degraded or minimal sensing. Building on KISS-IMU, our self-supervised inertial odometry framework, my current focus is proprioceptive state estimation for legged robots — explicitly modeling and correcting the environment-induced error in leg odometry using inertial, kinematic, and contact signals alone. I am an M.S.–Ph.D. integrated student at the SPARO Lab, Inha University, advised by Prof. Younggun Cho.

Highlights

- First-author papers at IEEE ICRA: KISS-IMU (self-supervised inertial odometry, 2026) and the PoLaRIS maritime detection & tracking dataset (co-first author, 2025).
- KISS-IMU: ICRA 2026 Best Paper Award Finalist on Robot Perception, and Winner of the Outstanding WiRA Student Paper Award, ICRA 2026.
- 2nd Place in two international robotics challenges: ICRA 2026 Hilti × Trimble SLAM (2nd/62 teams) and IROS 2026 TartanIMU (2nd/131 teams).
- Open-source author of KISS-IMU [★ **182**] and PoLaRIS [★ **63**].
- Reviewer for IEEE RA-L (2025–2026), IEEE ICRA (2026–2027), and IEEE/RSJ IROS (2026).

Field of Interest

My research interests include, but are not confined to:

- Self-supervised inertial learning and inertial odometry
- Robust and resilient state estimation under sensor degradation and limited supervision
- Weak and cross-modal supervision for inertial and motion learning

Education

Inha University — *Incheon, Republic of Korea*

M.S.–Ph.D. Integrated in Electrical and Computer Engineering — *Mar. 2024 - Current*

- Advised by Prof. Younggun Cho, Spatial AI and Robotics (SPARO) Lab

Inha University — *Incheon, Republic of Korea*

B.S. in Electrical Engineering — *Mar. 2020 - Feb. 2024*

Skills

Research Areas	SLAM, Self-supervised Inertial Learning, Robot Motion Analysis
Programming	C++, Python, MATLAB
Frameworks & Tools	ROS, Docker, Git, LaTeX
Languages	Korean (Native), English (Intermediate)

Research Project Experience

Motion-Environment Disentangled Representation Learning for Adaptive IMU Odometry — *SPARO Lab, Inha Univ.*

Principal Investigator, NRF Ph.D. Candidate Research Fellowship — *2026 – Present*

- Separates the robot's own motion from terrain and disturbance effects in IMU signals, so one inertial odometry framework adapts to new robots and environments without retraining.
- Keywords: Inertial odometry, Disentangled representation learning, Test-time adaptation

Cross-Embodiment Inertial Odometry — *SPARO Lab, Inha Univ.*

2nd Place (out of 131 teams), IROS 2026 TartanIMU Challenge — *2026*

- Built one shared-weight network that estimates body-frame velocity from raw 6-axis IMU across four platforms (car, quadruped, drone, hand-held), without platform labels at inference.
- Placed 2nd among 131 teams with over 89 test sequences (AVE 0.147 m/s, ATE_{20} 0.498 m).
- Keywords: Inertial odometry, Cross-embodiment learning, State-space models

Dual Fisheye Camera-based Visual SLAM *SPARO Lab, Inha Univ.*

2nd Place (out of 62 teams), ICRA 2026 Hilti x Trimble SLAM Challenge *2026*

- Built dual fisheye camera-based Visual SLAM pipeline for the challenge benchmark.
- Placed 2nd among 62 teams with an average trajectory error below 0.1 m across 25 sequences.
- Keywords: Visual SLAM, Dual fisheye camera, Localization accuracy

Publications

Bold denotes the author; underline denotes co-first authors or co-corresponding authors.

International Journal

1. Hogyun Kim, Juwon Kim, Dongjin Cho, **Jiwon Choi**, Geonmo Yang, Seokhwan Jeong, Jiyun Lee, Miryeong Park, Hyungtae Lim, and Younggun Cho, "SKiD-SLAM: Resource-Aware Distributed Multi-Robot LiDAR SLAM for Planetary Missions with Field Deployments at Analog Sites," *IEEE Trans. Field Robot. (T-FR)*, 2026.
2. Hogyun Kim, **Jiwon Choi**, Juwon Kim, Geonmo Yang, Seokhwan Jeong, Hyungtae Lim, and Younggun Cho, "Commerge: Communication-Efficient, Robust and Fast LiDAR Map Merging Framework for Multi-robot Coordination in Resource-constrained Scenarios," *Int. J. Robot. Res. (IJRR)*, 2026.
3. Hogyun Kim, **Jiwon Choi**, Taehu Sim, Giseop Kim, and Younggun Cho, "Narrowing Your FOV with SOLiD: Spatially Organized and Lightweight Global Descriptor for FOV-constrained LiDAR Place Recognition," *IEEE Robot. Automat. Lett. (RA-L) with ICRA*, 2024.

International Conference

1. Hogyun Kim, **Jiwon Choi**, Jungwoo Lee, and Younggun Cho, "RadLoc: Radar-based 3-DoF Global Localization via Fast, Robust, and Lightweight Spatial Descriptor Across Diverse Environmental Scenarios," *IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2026.
2. **Jiwon Choi**, Hogyun Kim, Geonmo Yang, Juhui Lee, and Younggun Cho, "KISS-IMU: Self-supervised Inertial Odometry with Motion-balanced Learning and Uncertainty-aware Inference," in *Proc. IEEE Int. Conf. Robot. Automat. (ICRA)*, 2026. **(Best Paper Award on Robot Perception, Finalist)**
3. **Jiwon Choi**, Dongjin Cho, Gihyeon Lee, Hogyun Kim, Geonmo Yang, Joowan Kim, and Younggun Cho, "PoLaRIS Dataset: A Maritime Object Detection and Tracking Dataset in Pohang Canal," in *Proc. IEEE Int. Conf. Robot. Automat. (ICRA)*, 2025.

Preprint

1. Hogyun Kim, **Jiwon Choi**, Juwon Kim, Geonmo Yang, Dongjin Cho, Hyungtae Lim, and Younggun Cho, "SKiD-SLAM: Robust, Lightweight, and Distributed Multi-Robot LiDAR SLAM in Resource-Constrained Field Environments," *arXiv preprint (ICRA Workshop on Field Robotics)*, 2025.

Honors & Awards

International

2026	**2nd Place (out of 131 teams),** TartanIMU Challenge, IROS 2026	*Pittsburgh, USA*
2026	**Winner,** Outstanding WiRA Student Paper Award, ICRA 2026	*Vienna, Austria*
2026	**Finalist,** Best Paper Award on Robot Perception, ICRA 2026	*Vienna, Austria*
2026	**2nd Place (out of 62 teams),** Hilti x Trimble SLAM Challenge, ICRA 2026	*Vienna, Austria*
2025	**Spotlight Talk,** ICRA 2025 3rd Workshop on Field Robotics	*Atlanta, USA*

Domestic

2026	**Finalist,** Qualcomm Innovation Fellowship Korea (QIFK) 2026	*Seoul, S.Korea*
2026	**Ph.D. Candidate Research Fellowship,** National Research Foundation of Korea (NRF)	*Daejeon, S.Korea*
2026	**Samsung HumanTech Paper Award (Silver Prize),** Samsung Electronics	*Seoul, S.Korea*
2026	**Graduate Research Award,** Inha University	*Incheon, S.Korea*
2025	**Excellent Research Award,** Korea Institute of ITS Conference	*Jeju, S.Korea*
2023	**Excellent Research Award (Undergraduate Division),** Institute of Control, Robotics and System	*Samcheok, S.Korea*
2022	**Alpha Project Competition (Grand Prize),** Inha University	*Incheon, S.Korea*
2022	**Excellent Research Award,** Korea Institute of ITS Conference	*Jeju, S.Korea*

2021 **Address-Based Industry Creation Idea Contest (Grand Prize),** Ministry of the Interior and Safety — *Sejong, S.Korea*

Patents

Granted

2025 **Domestic,** Method and System for Self-Driving Car-based Road Management System — *Incheon, S.Korea*

2024 **Domestic,** Method and System for Fire Response Based on Smart Unmanned Self-Driving Platform — *Incheon, S.Korea*

Application

2025 **International,** Lightweight Robot Localization for LiDARs with Limited FOV — *Incheon, S.Korea*

2024 **Domestic,** Lightweight Robot Localization for LiDARs with Limited FOV — *Incheon, S.Korea*

2023 **Domestic,** A Method on the Efficient Multi-Floor Navigation Using Visual Marker and Floor Plan Map — *Incheon, S.Korea*

Academic Activities

Reviewer

- IEEE Robotics and Automation Letters **(RA-L 2025, 2026)**
- IEEE International Conference on Robotics and Automation **(ICRA 2026, 2027)**
- IEEE/RSJ International Conference on Intelligent Robots and Systems **(IROS 2026)**
- International Conference on Control, Automation and Systems **(ICCAS 2026)**